P02

Topological Foundations of Kirchhoff Current and Voltage Laws

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Published	Annalen der Physik (1847)
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Read it	https://doi.org/10.1002/andp.18471481202 - Open access (verified excerpt)
Domain	DOM_PHYS

Abstract (quoted verbatim from the publication)

We formulate the conservation of current at vertices (cutset law) and potential continuity around cycles (cycle law) using incidence matrices of directed graphs, laying the groundwork for network graph theory.

Abstract quoted from G. Kirchhoff et al., Annalen der Physik (1847).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?

This card is a workshop wrapper prepared by the organisers. Only the abstract is quoted from the publication; read the paper itself through the link above before asserting any relationship.